THE COMPLETE UNIVERSAL AI PROMPT ENGINEERING HANDBOOK

A comprehensive guideline for GPT-5.6 Luna, Gemini 3.8 Flash and other advanced AI models

Purpose: To maximize the quality, accuracy, efficiency, creativity, and reliability of AI-generated responses across different domains and tasks.

PART 1 — THE FUNDAMENTAL PHILOSOPHY OF PROMPT ENGINEERING

1. What makes a prompt effective?

An effective prompt communicates five essential things:

1. **Context:** What does the AI need to know?
2. **Objective:** What exactly must the AI accomplish?
3. **Instructions:** How should it approach the task?
4. **Constraints:** What must it follow or avoid?
5. **Success criteria:** How will the final result be evaluated?

A prompt should give the AI enough information to understand the task without overwhelming it with unnecessary instructions.

The objective is not to write the longest possible prompt. The objective is to provide the most useful information in the clearest possible way.

2. The universal prompting formula

Use this formula as the foundation:

ROLE + CONTEXT + OBJECTIVE + REQUIREMENTS + CONSTRAINTS + EXECUTION + VERIFICATION + OUTPUT

Component	Purpose
Role	Establish the expertise and responsibilities expected
Context	Explain the background and relevant information
Objective	Define the exact task

Requirements	Specify what must be accomplished
Constraints	Establish boundaries and restrictions
Execution	Explain the expected approach
Verification	Define how accuracy and completeness are checked
Output	Specify the final presentation format

Not every simple prompt needs all eight components. Use the components that materially affect the task.

3. Think in terms of outcomes, not just instructions

Do not unnecessarily prescribe every action the AI must take.

Instead:

- Define the desired outcome.
- Explain the important constraints.
- Specify the quality standard.
- Allow the AI to choose an appropriate method.
- Require verification of the result.

Give detailed procedural instructions only when the process itself matters.

4. Prioritize information correctly

Place the most important information first:

1. Critical objective.
2. Mandatory requirements.
3. Essential context.
4. Important restrictions.
5. Supporting information.
6. Output requirements.

For long prompts, provide the supporting material first and place the final specific task at the end, with a clear instruction connecting the two.

PART 2 — ROLE DEFINITION AND PERSONA DESIGN

1. Give the AI an appropriate role

A role helps establish the perspective and expertise expected from the model.

Examples:

Research:
"You are an experienced research analyst specializing in evidence-based research, source evaluation, and factual verification."

Programming:
"You are a senior software engineer and system architect specializing in reliable software development, debugging, security, and optimization."

Mathematics:
"You are an expert mathematics educator with strong problem-solving and analytical skills."

Business:
"You are a business strategist specializing in market research, financial analysis, competitive positioning, and practical execution."

Design:
"You are an experienced UI/UX designer specializing in usability, visual hierarchy, accessibility, and modern interface design."

2. Define the role's responsibilities

Do not stop at assigning a title.

Specify what the role must achieve:

- Analyze the problem.
- Apply relevant expertise.
- Evaluate alternatives.
- Identify risks.
- Produce practical recommendations.
- Verify important conclusions.
- Communicate findings clearly.

3. Avoid excessive role descriptions

A prompt does not become more powerful simply because the AI is assigned ten professional titles.

Use one primary role and add supporting expertise only when relevant.

4. Define the audience

Explain who the final response is intended for.

Examples:

- Complete beginner.
- University student.
- Software developer.
- Technical expert.
- Business owner.
- General audience.
- Examination evaluator.

The audience determines the terminology, complexity, depth, and explanation style.

PART 3 — CONTEXT ENGINEERING

Context is one of the most important factors affecting output quality.

1. Provide relevant background information

Include:

- Project background.
- Previous decisions.
- Existing work.
- Important preferences.
- Available resources.
- Intended audience.
- Technical environment.
- Relevant limitations.
- Desired outcomes.

The AI should not be expected to infer important facts that have not been supplied.

2. Establish a source of truth

When providing documents, files, screenshots, datasets, or specifications, explain which material should be treated as authoritative.

Example:

"The attached project specification is the primary source of truth. Follow its requirements. If you use external information to supplement it, clearly distinguish external information from the supplied material."

For source-dependent tasks:

- Do not invent missing details.
- Do not silently change source terminology.
- Do not assume that an incomplete document contains information that is not present.

- Clearly identify contradictions.
- Explain when the available material is insufficient.

3. Provide context in a structured form

Use headings, lists, tables, or clearly separated sections.

Example:

```
PROJECT BACKGROUND:
[Background]

CURRENT SITUATION:
[Current status]

PREVIOUS DECISIONS:
[Decisions already made]

AVAILABLE RESOURCES:
[Resources]

RESTRICTIONS:
[Limitations]

EXPECTED RESULT:
[Outcome]
```

4. Distinguish facts, assumptions, and opinions

Instruct the AI to separate:

- **Facts:** Information supported by reliable evidence.
- **Assumptions:** Information used because something is unknown.
- **Inferences:** Conclusions derived from available information.
- **Opinions:** Subjective evaluations.
- **Recommendations:** Suggested actions.

Never present an assumption as a confirmed fact.

5. Maintain context across conversations

For ongoing projects:

- Refer to previous decisions.
- Preserve established requirements.
- Reuse existing terminology.
- Track completed and incomplete work.
- Identify changes to previous instructions.
- Do not repeat already settled questions unnecessarily.

If the relevant earlier information is unavailable, ask for the missing context rather than inventing it.

6. Avoid context overload

Do not provide every piece of information you have.

Include information that affects:

- The objective.
- The implementation.
- The constraints.
- The correctness of the answer.
- The expected output.

Remove unrelated details and duplicate content.

PART 4 — OBJECTIVE AND TASK DEFINITION

1. Define the task precisely

A task should clearly communicate what the AI must produce.

Examples:

Weak:
"Explain artificial intelligence."

Better:
"Explain artificial intelligence to a first-year engineering student. Cover its definition, major approaches, practical applications, limitations, and examples. Use clear language and organize the explanation into logical sections."

2. Define the expected outcome

Specify what successful completion looks like.

For example:

- A fully functioning application.
- A research report supported by credible sources.
- A detailed but concise study guide.
- A presentation with seven slides.
- A validated calculation.

- A comparison table with a recommendation.

3. Separate multiple tasks

If a prompt contains several objectives, list them separately.

Example:

```
TASK 1: Research the topic.
TASK 2: Analyze the information.
TASK 3: Compare available alternatives.
TASK 4: Recommend the best approach.
TASK 5: Present the final result.
```

This reduces the risk of overlooking one part.

4. Define the scope

Specify:

- What is included.
- What is excluded.
- How detailed the answer should be.
- Whether external research is permitted.
- Whether the AI should make recommendations.
- Whether the AI should execute actions or only explain them.

Avoid scope expansion without permission.

5. Establish task priorities

When multiple objectives compete, specify which matters most.

For example:

1. Accuracy.
2. Compliance with requirements.
3. Completeness.
4. Practical usefulness.
5. Efficiency.
6. Creativity.
7. Presentation.

The order may change depending on the task.

PART 5 — INSTRUCTION DESIGN AND CLARITY

1. Use direct language

Prefer:

- Analyze.
- Compare.
- Calculate.
- Verify.
- Explain.
- Implement.
- Summarize.
- Evaluate.
- Recommend.
- Generate.

Avoid vague instructions such as:

- Make it amazing.
- Do something advanced.
- Make it very intelligent.
- Make it perfect.
- Make it world-class.

If you use subjective terms, define what they mean in measurable or observable terms.

2. Be specific about requirements

Instead of:

"Make a detailed presentation."

Write:

"Create a seven-slide presentation. Each slide must have a clear title, a logical information hierarchy, concise explanations, and relevant evidence. Avoid overcrowding and ensure that the presentation has a coherent narrative."

3. Use one instruction for one purpose

Avoid combining several unrelated instructions into one complicated sentence.

Separate:

- Content requirements.
- Formatting requirements.
- Technical restrictions.
- Verification requirements.
- Delivery requirements.

4. Avoid contradictory instructions

Check for contradictions before submitting the prompt.

Examples:

- "Give a short answer" and "Explain every detail."
- "Use only the supplied document" and "Include current external research."
- "Do not change existing functionality" and "Replace the entire application."
- "Provide exactly five options" and "Include every possible option."

When requirements conflict, instruct the AI to identify the conflict and seek clarification.

5. Define mandatory and optional requirements

Use labels:

MUST: Non-negotiable requirement.

SHOULD: Strong preference.

MAY: Optional improvement.

MUST NOT: Prohibited action.

This makes instruction priorities easier to understand.

6. Use positive instructions alongside restrictions

Instead of only stating what the AI must not do, explain what it should do instead.

Example:

"Do not invent sources. Use credible, verifiable sources and explicitly state when supporting evidence is unavailable."

PART 6 — REASONING AND PROBLEM-SOLVING ENHANCEMENT

1. Encourage careful analysis

Ask the AI to:

- Understand the complete problem.
- Identify the important variables.
- Break complex tasks into smaller parts.
- Examine relevant dependencies.
- Consider alternative approaches.
- Evaluate trade-offs.
- Identify possible failure scenarios.
- Verify the final result.

2. Use decomposition for complex tasks

Break a large objective into manageable stages.

Example:

```
STAGE 1: Understand the problem.
STAGE 2: Identify requirements.
STAGE 3: Research or analyze available information.
STAGE 4: Develop possible approaches.
STAGE 5: Evaluate alternatives.
STAGE 6: Select the most suitable solution.
STAGE 7: Execute the solution.
STAGE 8: Verify the result.
STAGE 9: Deliver the final output.
```

Do not require every stage for a simple question.

3. Request alternative solutions

For difficult problems, ask the AI to consider multiple approaches.

Example:

"Evaluate at least two practical approaches. Compare their accuracy, complexity, cost, efficiency, limitations, and suitability. Select the best approach and justify the decision."

Do not require alternatives when the task has only one valid method.

4. Encourage critical thinking

Ask the AI to challenge its own conclusions.

It should consider:

- What could be wrong?

- What assumptions are being made?
- Is there another interpretation?
- Is important information missing?
- Are there exceptions?
- Does the conclusion actually follow from the evidence?

5. Use examples and test cases

Examples help clarify complex requirements.

Provide:

- Normal cases.
- Boundary cases.
- Invalid cases.
- Expected outputs.
- Special scenarios.

6. Request conclusions, not hidden reasoning transcripts

Ask the model to perform the necessary analysis internally and present:

- The conclusion.
- Important assumptions.
- Key calculations.
- Relevant evidence.
- Concise reasoning or justification.
- Verification results.

Do not require private internal chain-of-thought. A concise, auditable explanation is more useful.

PART 7 — THE COMPULSORY DOUBLE-VERIFICATION PROTOCOL

<THIS IS A MANDATORY SECTION. INCLUDE IT IN EVERY IMPORTANT PROMPT.>

1. The central instruction

Every substantial response must undergo two separate review passes before submission.

First review — Correctness and completeness

Check:

- Did the AI understand the task correctly?
- Are all requirements addressed?
- Are the facts accurate?
- Are calculations correct?
- Is the logic valid?
- Are assumptions reasonable?
- Are sources credible?
- Are there missing components?
- Are constraints followed?

Second review — Independent quality and error detection

Re-examine the completed answer from a critical perspective.

Check:

- Are there hidden mistakes?
- Are there contradictions?
- Are any important details overlooked?
- Are there unsupported statements?
- Are there incorrect calculations?
- Is the output properly formatted?
- Does the final answer actually satisfy the objective?
- Can any part be improved without violating requirements?

Correct every identified issue before submission.

2. Use the following compulsory prompt instruction

```
COMPULSORY DOUBLE-VERIFICATION INSTRUCTION:

Before submitting your final response, perform two
separate review passes.

FIRST REVIEW:
Carefully verify the accuracy, relevance, logic,
completeness, calculations, assumptions, and compliance
of the entire response against the original requirements.

SECOND REVIEW:
Re-examine the complete result independently.
Look specifically for overlooked requirements,
factual mistakes, logical inconsistencies, incorrect
calculations, missing information, unsupported claims,
formatting problems, and avoidable errors.

Correct every identified issue before finalizing.

Cross-check important information using reliable
sources, independent calculations, available tools,
or actual execution wherever appropriate.

Do not invent information, sources, test results,
or verification claims.
```

```
If something cannot be verified, explicitly disclose
that limitation.

Do not mark the task complete until the final output
has been checked against the original objective
and every mandatory requirement.
```

3. Verification must match the task

Different tasks require different verification methods.

Task	Appropriate verification
Mathematics	Recalculate independently and check the method
Research	Cross-check facts against credible sources
Programming	Run relevant tests and inspect actual behaviour
Data analysis	Validate data, calculations, and conclusions
Writing	Check grammar, structure, tone, and factual consistency
Presentations	Check slide content, layout, readability, and consistency
Image generation	Check composition, text, proportions, and requested details
Business planning	Check assumptions, feasibility, costs, and risks
Academic notes	Compare against supplied material and syllabus

4. Never confuse reviewing with testing

Reading code twice does not prove it works.

Checking a calculation twice using the same incorrect formula does not prove it is correct.

A meaningful verification process uses the most appropriate independent check available.

Examples:

- Recalculate a mathematical answer using a different method.
- Run code using actual inputs.
- Compare research claims against primary sources.
- Check a database operation against actual records.
- Verify a generated presentation against the original slide requirements.

5. Establish a completion standard

The AI must distinguish between:

- **Verified:** Checked using appropriate evidence or testing.
- **Partially verified:** Some, but not all, relevant aspects were checked.
- **Unverified:** Not independently checked.
- **Assumed:** Based on a stated assumption.

- **Uncertain:** Evidence is insufficient or conflicting.

This prevents false confidence.

PART 8 — ACCURACY, FACT-CHECKING AND RESEARCH

1. Prioritize factual accuracy

For factual tasks:

- Use reliable sources.
- Prefer primary sources.
- Check publication dates.
- Check whether information is current.
- Verify important figures.
- Identify conflicting information.
- Avoid unsupported claims.
- Distinguish facts from interpretation.

2. Use a source hierarchy

A useful general hierarchy is:

1. Official government and regulatory sources.
2. Official technical documentation.
3. Original research papers and primary studies.
4. Reputable academic institutions.
5. Established industry publications.
6. Credible news organizations.
7. Expert commentary.
8. Community discussions and informal sources.

The most suitable source depends on the subject. For example, official software documentation is usually preferable for API specifications, while original research is more appropriate for scientific findings.

3. Request citations where necessary

For research-intensive tasks, require:

- Source title.
- Publishing organization.
- Publication date where available.
- Direct source link.

- Supporting evidence.
- Relevant limitations.

Citations must support the specific claims they accompany.

4. Verify recent information

When a question concerns current information, explicitly instruct the AI to research it.

Examples:

- Current software versions.
- API changes.
- Government policies.
- Admission deadlines.
- Market prices.
- Recent scientific developments.
- Current regulations.
- Company announcements.

A model's stored knowledge should not be treated as a substitute for checking current sources.

5. Handle conflicting sources

If sources disagree:

1. Identify the disagreement.
2. Compare source authority.
3. Compare publication dates.
4. Examine the original evidence.
5. Explain the most credible interpretation.
6. Disclose unresolved uncertainty.

Do not silently choose the source that supports a preferred conclusion.

6. Prevent hallucinations

Use this instruction:

"Do not fabricate facts, sources, quotations, statistics, citations, technical specifications, or references. If the required information is unavailable or cannot be verified, state that clearly rather than guessing."

7. Separate source-based and model-generated information

When working from documents, distinguish:

- Directly supported content.
- External research.
- General background knowledge.
- Inferences.
- Recommendations.

This is particularly important in academic, legal, scientific, and technical tasks.

PART 9 — CREATIVITY AND INNOVATION

1. Encourage meaningful creativity

Creativity should improve the outcome, not merely make it different.

Ask the AI to:

- Explore useful alternatives.
- Identify overlooked opportunities.
- Suggest unconventional but practical approaches.
- Improve the structure or presentation.
- Consider user experience.
- Find simpler solutions.
- Identify opportunities for automation.
- Think about future scalability.

2. Use creativity within boundaries

Specify:

- What can be changed.
- What must remain unchanged.
- Which improvements are permitted.
- Which additions are optional.
- What budget or resource restrictions apply.

3. Request innovative alternatives

Example:

"After developing the core solution, identify three genuinely useful improvements that could make it more effective. Evaluate their feasibility, cost, complexity, and practical value. Do not add unnecessary features."

4. Avoid generic creativity

Do not accept suggestions that are:

- Unrelated to the objective.
- Impractical.
- Expensive without justification.
- Technically infeasible.
- Unnecessarily complicated.
- Repetitions of existing features.

PART 10 — EFFICIENCY AND RESPONSE OPTIMIZATION

1. Optimize prompt length

A longer prompt is not necessarily a better prompt.

Keep:

- Essential context.
- Mandatory requirements.
- Relevant constraints.
- Verification instructions.
- Expected output.

Remove:

- Repeated instructions.
- Unnecessary descriptions.
- Unrelated background information.
- Duplicate examples.
- Excessive role descriptions.

2. Specify response length

Use explicit requirements:

Short: Direct answer with essential information only.

Medium: Explanation with relevant examples and important details.

Detailed: Comprehensive explanation covering major aspects, examples, limitations, and recommendations.

Exhaustive: In-depth treatment of all relevant areas, with structured sections and detailed analysis.

Do not ask for an exhaustive response when a short answer is sufficient.

3. Avoid unnecessary clarification

Tell the AI:

"If minor information is missing, make a reasonable assumption, state it briefly, and proceed. If missing information materially affects correctness, safety, cost, or the final outcome, ask a focused clarification question."

4. Avoid unnecessary repetition

If a requirement is already clearly stated, do not repeat it in five different sections.

A single well-defined instruction is usually more effective.

5. Use tools selectively

Tools should be used when they materially improve the result.

Examples:

- Web research for current information.
- Code execution for calculations and programming.
- File analysis for supplied documents.
- Data analysis for large datasets.
- Image analysis for visual references.
- Connected applications for authorized external tasks.

Do not use tools merely to appear thorough.

6. Balance quality, time, and resources

Specify whether the task prioritizes:

- Maximum accuracy.
- Minimum latency.
- Lower cost.
- Detailed analysis.
- Practical usability.
- A balanced combination.

The model should use an appropriate level of effort.

PART 11 — MULTIMODAL PROMPTING

AI models may receive text, images, documents, audio, video, or structured data.

The prompt must explain how each source should be used.

1. Image prompting

When providing images, specify:

- What the image represents.
- Which elements should be examined.
- Which details must remain unchanged.
- What should be generated or modified.
- Desired composition.
- Aspect ratio.
- Resolution or output dimensions.
- Style and visual references.
- Text that must appear.
- Elements that must not appear.

For image editing, explicitly identify the areas that can change and the areas that must remain identical.

2. Document prompting

For documents:

- Identify the relevant file.
- Explain the requested operation.
- Specify whether the entire document or selected sections matter.
- Preserve required terminology.
- Specify the desired output format.
- Request citations or page references when necessary.

3. Audio prompting

Specify:

- Whether the task is transcription, summarization, analysis, or generation.
- Required language.
- Speaker identification requirements.
- Whether timestamps are needed.
- Whether exact wording must be preserved.
- Whether background noise or unclear sections should be marked.

4. Video prompting

Specify:

- Relevant timestamps.
- Events to identify.
- Whether visual and audio information should be combined.
- Whether chronological ordering matters.
- Whether a detailed or concise summary is required.

5. Multiple sources

When several files or media sources are supplied, assign each one a purpose.

Example:

"Use Document A as the primary technical specification, Image B as the design reference, and Spreadsheet C as the source of numerical data."

This reduces source confusion.

PART 12 — OUTPUT FORMATTING AND PRESENTATION

A correct answer can still be poor if it is difficult to understand or use.

1. Specify the output format

Choose the appropriate format:

- Paragraphs.
- Bullet points.
- Numbered steps.
- Tables.
- Markdown.
- JSON.
- Code blocks.
- Reports.
- Presentations.
- Checklists.
- Structured documents.

2. Define the structure

Example:

```
1. Executive Summary
2. Background
3. Analysis
4. Findings
5. Recommendations
6. Limitations
7. Conclusion
```

3. Define formatting requirements

Specify:

- Headings.
- Section order.
- Table columns.
- Number of items.
- Word limits.
- Language.
- Tone.
- Technical depth.
- Citation format.

4. Prioritize readability

The final response should:

- Be logically organized.
- Avoid unnecessary repetition.
- Use consistent terminology.
- Explain technical terms where necessary.
- Highlight important conclusions.
- Present information in a usable format.

5. Match the output to the user's purpose

For example:

- A student may need detailed explanations.
- A developer may need complete code and setup instructions.
- A judge may need measurable results and evidence.
- A manager may need concise recommendations.
- A researcher may need sources and limitations.

PART 13 — TOOL USE, AUTONOMY AND EXECUTION

1. Define what the AI is allowed to do

For agentic or tool-enabled systems, specify the permitted actions.

Examples:

- Read files.
- Search the web.
- Analyze datasets.
- Execute code.
- Modify project files.
- Run tests.
- Create documents.
- Interact with connected applications.

2. Define approval boundaries

Use a clear policy:

- Allow ordinary, non-destructive actions that are necessary for the requested task.
- Allow in-scope file inspection, code editing, and testing when implementation is requested.
- Ask before destructive operations.
- Ask before external writes or irreversible actions.
- Ask before purchases or financial commitments.
- Ask before expanding the project beyond its requested scope.

3. Require evidence from tool use

When the AI uses tools:

- Distinguish tool output from assumptions.
- Check tool errors.
- Verify important returned values.
- Avoid inventing successful actions.
- Report incomplete operations.

4. Prevent unnecessary tool loops

Instruct the AI to:

- Use the most relevant tool.
- Avoid repeated searches without a reason.

- Stop when sufficient evidence has been obtained.
- Change approach if a tool repeatedly fails.
- Report when a required tool is unavailable.

PART 14 — MODEL-SPECIFIC OPTIMIZATION

The general guidelines above apply to both models. The following settings and prompting strategies are specific to their respective model families.

A. GPT-5.6 LUNA

API model ID: `gpt-5.6-luna`

Recommended prompting approach

1. Define the intended outcome clearly.
2. Provide relevant context.
3. State hard constraints explicitly.
4. Define success criteria.
5. Explain what must be preserved.
6. Establish autonomy and approval boundaries.
7. Avoid repeated or unnecessarily restrictive instructions.
8. Let the model choose an appropriate approach within the requirements.
9. Encourage tool use when it materially improves accuracy.
10. Require validation and honest reporting.

Reasoning effort

The GPT-5.6 model family supports these API reasoning-effort settings:

Setting	Suggested use
`none`	Simple tasks with low reasoning requirements
`low`	Routine work and quick responses
`medium`	General-purpose balanced tasks
`high`	Complex analysis and problem-solving
`xhigh`	Difficult multi-step tasks
`max`	Particularly demanding, quality-first tasks

These are suggested starting points. The best setting depends on the workload.

GPT-5.6 also supports standard and pro reasoning modes in the Responses API. These are separate from reasoning effort.

Practical recommendation:

- Use medium for ordinary tasks.
- Use high or xhigh for complex work when additional reasoning improves results.
- Reserve max or pro mode for demanding tasks where additional latency and token usage are acceptable.

GPT-5.6 prompting principle

Do not micromanage every step unnecessarily. Provide the goal, context, constraints, permissions, and success criteria. Allow the model to carry out safe, in-scope work and verify its results.

B. GEMINI 3.8 FLASH

API model ID: `gemini-3.8-flash`

Recommended prompting approach

1. Be direct and precise.
2. Use clear headings or XML-style tags.
3. Separate context from instructions.
4. Define ambiguous terms.
5. State critical requirements near the beginning.
6. Explain the expected response depth.
7. Use examples for unusual requirements.
8. Provide large context before the final task.
9. Clearly reference images, documents, audio, and video.
10. Request verification of important conclusions.

Thinking level

Gemini 3.8 Flash supports:

Setting	Suggested use
`low`	Quick, straightforward tasks
`medium`	General tasks and substantial coding work
`high`	Difficult reasoning, mathematics, and complex multi-step tasks

`medium` is the default. `minimal` is unsupported.

Important API considerations

For Gemini 3.8 Flash:

- Use the documented `thinking_level` configuration.
- Do not rely on unsupported legacy sampling settings.
- Check the current API documentation for parameter compatibility.

- Ensure function-call and function-response records match correctly.
- Place multimodal assets in the appropriate request payload.
- Follow the API's conversation-history requirements.

These are API implementation details, not instructions that must be copied into every ordinary ChatGPT or Gemini chat prompt.

Gemini prompting principle

Use concise, direct, consistently structured instructions. Make the objective and constraints unmistakable. Explicitly request detailed output when needed rather than assuming the model will always provide it.

PART 15 — SPECIALIZED PROMPTING STRATEGIES

1. Research prompts

Include:

- Research objective.
- Scope.
- Time period.
- Geographic coverage.
- Source requirements.
- Important questions.
- Comparison criteria.
- Citation requirements.
- Expected report structure.

Require the AI to identify evidence gaps and conflicting sources.

2. Academic learning prompts

Include:

- Subject.
- Topic.
- Current knowledge level.
- Syllabus.
- Learning objectives.
- Desired depth.
- Examples.
- Important terminology.
- Exam relevance.

- Expected notes format.

Ask the AI to explain concepts accurately and build understanding progressively.

3. Mathematical problem-solving prompts

Include:

- Complete problem statement.
- Given values.
- Required result.
- Units.
- Applicable conditions.
- Required method, if any.
- Expected precision.

Require independent recalculation, unit checks, and verification of the final answer.

4. Business strategy prompts

Include:

- Business idea.
- Market.
- Target customers.
- Budget.
- Resources.
- Competition.
- Operational constraints.
- Revenue model.
- Risks.
- Time horizon.

Require realistic assumptions, cost analysis, feasibility, and practical recommendations.

5. Data analysis prompts

Include:

- Dataset description.
- Data source.
- Column definitions.
- Objective.
- Expected analysis.
- Missing-data policy.
- Statistical requirements.
- Visualizations.
- Output format.

Require validation of data quality, calculations, and interpretations.

6. Writing prompts

Include:

- Purpose.
- Audience.
- Tone.
- Length.
- Main message.
- Required details.
- Prohibited content.
- Output format.

Ask the AI to check factual consistency, clarity, grammar, and naturalness.

7. Image-generation prompts

Include:

- Main subject.
- Intended purpose.
- Composition.
- Perspective.
- Environment.
- Style.
- Lighting.
- Colours.
- Aspect ratio.
- Required text.
- Elements to exclude.

For precise visual work, distinguish between essential details and creative freedom.

8. Decision-making prompts

Include:

- Decision to be made.
- Available options.
- Objectives.
- Constraints.
- Evaluation criteria.
- Risks.
- Time horizon.

Request a comparison, trade-off analysis, and justified recommendation.

PART 16 — UNIVERSAL MASTER PROMPT

The following template combines the most important principles in one reusable prompt.

Use it as the foundation for substantial tasks. Customize it to the objective rather than blindly including every section.

```
# ROLE AND RESPONSIBILITY

Act as a highly capable expert in the domain relevant
to this task.

Apply strong analytical thinking, domain knowledge,
accuracy, creativity, and practical problem-solving.

Your responsibility is to deliver a complete,
reliable, relevant, and high-quality result.

# 1. CONTEXT

[Provide all relevant background information,
documents, files, references, and previous decisions.]

Treat the supplied authoritative materials as the
source of truth wherever specified.

Do not invent missing information.

# 2. OBJECTIVE

[Explain exactly what needs to be accomplished.]

The intended outcome is:

[Describe the desired result.]

# 3. REQUIREMENTS

[List all mandatory requirements.]

Clearly distinguish mandatory requirements,
preferences, and optional improvements.

# 4. CONSTRAINTS

[List technical, practical, financial, time-related,
content-related, and other restrictions.]

Do not violate mandatory restrictions.

Do not introduce unrelated changes or unnecessary
complexity.

# 5. EXECUTION
```

Understand the complete task before producing
the final result.

Break complex problems into manageable parts.

Identify relevant dependencies and risks.

Consider appropriate alternative approaches.

Choose the approach that best satisfies the
requirements.

Use available tools, documents, and evidence
when they materially improve the result.

6. ACCURACY AND SOURCE REQUIREMENTS

Prioritize correctness over speed or unsupported
confidence.

Do not fabricate facts, sources, calculations,
references, or technical details.

Verify current information using reliable sources
when the task requires up-to-date information.

Clearly distinguish facts, assumptions,
inferences, and recommendations.

7. CREATIVITY AND OPTIMIZATION

Identify practical improvements and useful
alternatives where appropriate.

Prioritize simplicity, effectiveness, efficiency,
and real-world feasibility.

Do not add unnecessary features or complexity.

8. OUTPUT REQUIREMENTS

Provide the result in the following format:

[Specify the required structure, length,
language, tone, and presentation.]

Ensure that the result is complete, clear,
organized, and directly usable.

9. COMPULSORY DOUBLE-VERIFICATION

Before submitting the final result, perform
two separate review passes.

FIRST REVIEW:
Check the complete response for accuracy,
relevance, completeness, logical consistency,
correct calculations, correct interpretation,
and compliance with every mandatory requirement.

```
SECOND REVIEW:
Re-examine the entire result independently.
Look for overlooked details, hidden mistakes,
unsupported claims, contradictions, incorrect
assumptions, formatting problems, and areas
that require correction.

Cross-check important facts, calculations,
and conclusions using suitable independent
methods, reliable sources, or actual tests.

Correct every identified issue before submitting.

Never claim that something has been tested,
executed, researched, or verified unless
that action was actually performed.

If something cannot be verified, clearly
disclose the limitation.

# 10. FINAL COMPLETION STANDARD

The task is complete only when the result:

- Addresses the actual objective.
- Satisfies all mandatory requirements.
- Respects all constraints.
- Is accurate and logically consistent.
- Has undergone both review passes.
- Has been verified to the extent possible.
- Clearly discloses remaining uncertainties.

Deliver the best practical result without
unnecessary delays, unsupported assumptions,
or avoidable mistakes.
```

PART 17 — THE 40 GOLDEN RULES OF AI PROMPTING

Understanding and context

1. Always define the objective clearly.
2. Provide relevant context.
3. Identify the intended audience.
4. Establish the source of truth.
5. Preserve important previous decisions.
6. Separate facts from assumptions.

Instructions and structure

7. Use direct language.
8. Organize complex prompts into sections.

9. State mandatory requirements explicitly.
10. Distinguish requirements from preferences.
11. Avoid contradictory instructions.
12. Avoid unnecessary repetition.
13. Define the scope.
14. Specify the expected output.

Reasoning and problem-solving

15. Encourage careful analysis.
16. Break complex tasks into manageable components.
17. Consider alternatives when useful.
18. Examine risks and edge cases.
19. Challenge assumptions.
20. Ask for evidence-based conclusions.

Accuracy and verification

21. Prioritize factual accuracy.
22. Use reliable sources when necessary.
23. Check current information.
24. Verify important calculations.
25. Cross-check important conclusions independently.
26. Perform two separate review passes.
27. Correct identified errors.
28. Never accept fabricated verification claims.

Efficiency and creativity

29. Keep prompts concise but complete.
30. Avoid unnecessary clarification.
31. Use tools selectively.
32. Avoid overcomplicating simple tasks.
33. Encourage practical creativity.
34. Optimize according to actual requirements.

Final output

35. Match the response to the user's purpose.
36. Use clear formatting.
37. Include relevant limitations.
38. Provide actionable results.
39. Confirm that every mandatory requirement is addressed.
40. Never declare completion without appropriate verification.

PART 18 — FINAL AI PROMPT QUALITY SCORECARD

Use this scorecard to assess the quality of an important prompt and its resulting answer.

Criterion	Evaluation
Clarity	Is the task unambiguous?
Context	Does the AI have sufficient information?
Relevance	Does the answer directly address the objective?
Accuracy	Are facts and conclusions correct?
Completeness	Are all mandatory requirements covered?
Reasoning	Is the problem solved properly?
Creativity	Are useful improvements considered?
Efficiency	Is the approach practical and optimized?
Reliability	Are assumptions and uncertainties handled correctly?
Verification	Has the result been checked twice?
Presentation	Is the final answer clear and usable?

Suggested rating:

- 9–10: Excellent.
- 7–8: Good.
- 5–6: Acceptable, needs improvement.
- 3–4: Poor.
- 0–2: Unacceptable.

A high average must not hide a critical failure. A fabricated citation, incorrect core calculation, violation of a mandatory restriction, or unverified claim of successful execution can make the entire result unacceptable.

FINAL PRINCIPLE: THE STANDARD FOR EXCELLENT AI OUTPUT

A high-quality AI response should satisfy all of the following:

UNDERSTAND: Correctly interpret the objective and context.

ANALYZE: Identify the important requirements, dependencies, and challenges.

REASON: Evaluate suitable approaches and avoid unsupported assumptions.

CREATE: Produce a useful, relevant, and well-structured result.

OPTIMIZE: Avoid unnecessary complexity and resource consumption.

VERIFY: Check the result twice using appropriate methods.

CORRECT: Fix identified errors and recheck affected areas.

DELIVER: Present a complete, honest, and usable final output.

The ultimate goal is not simply to make AI generate more content or use more reasoning. It is to make AI produce the **most accurate, relevant, efficient, creative, and dependable result possible for the actual task.**

The compulsory rule: Never submit the final result without reviewing it twice, cross-checking important details, correcting identified mistakes, and honestly disclosing anything that remains unverified.